

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in the KDD process and justify the role of data preprocessing.

Data Mining:

Data mining is the process of extracting useful patterns, relationships, and knowledge from large datasets.

Any Two Basic Data Mining Tasks:

- Classification
- Clustering

Steps in KDD (Knowledge Discovery in Databases):

1. Data Cleaning
2. Data Integration
3. Data Selection
4. Data Transformation
5. Data Mining
6. Pattern Evaluation
7. Knowledge Presentation

Role of Data Preprocessing:

- Removes missing values and noise.
- Integrates data from different sources.
- Transforms data into suitable formats.
- Reduces data size while preserving important information.
- Improves the accuracy and efficiency of data mining.

Q2. What is Data Cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data.

Data Cleaning:

Data cleaning is the process of detecting and correcting incomplete, inconsistent, or inaccurate data.

Common Data Quality Issues:

- Missing values
- Noisy data

Handling Missing Values:

- Ignore the record (if very few).
- Replace with mean, median, or mode.
- Predict using regression or machine learning.

Example:

Age = ?, replace with average age = 30 years.

Handling Noisy Data:

- Binning
- Regression
- Clustering

Example:

Marks = 45, 46, 47, 95 → Smooth using binning.

Q3. Define Data Integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled.

Data Integration:

Data integration combines data from multiple databases into one consistent dataset.

Challenges:

- Data redundancy
- Data inconsistency

Handling Redundancy:

- Remove duplicate tuples.
- Perform correlation analysis.

Handling Inconsistency:

- Standardize formats.
- Apply integrity constraints.
- Resolve conflicting values.

Q4. What is Covariance Analysis? List the two covariance formulas.

Covariance Analysis:

Covariance measures how two variables change together.

Population Covariance

$$\text{Cov}(X,Y) = \frac{\sum (X - \mu_X)(Y - \mu_Y)}{N}$$

Sample Covariance

$$\text{Cov}(X,Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n-1}$$

Interpretation:

- Positive covariance → variables move together.
- Negative covariance → variables move in opposite directions.
- Zero covariance → no linear relationship.

Q5. Suppose two stocks A and B have the following values:

A = (2, 3, 5, 4, 6)

B = (5, 8, 10, 11, 14)

Answer:

Both stock prices generally increase together.

- Covariance is **positive**.
- Therefore, the stocks **rise together**.
- They are **not independent** because both are influenced by the same industry trend.

Q6. Explain the architecture of a typical data mining system.

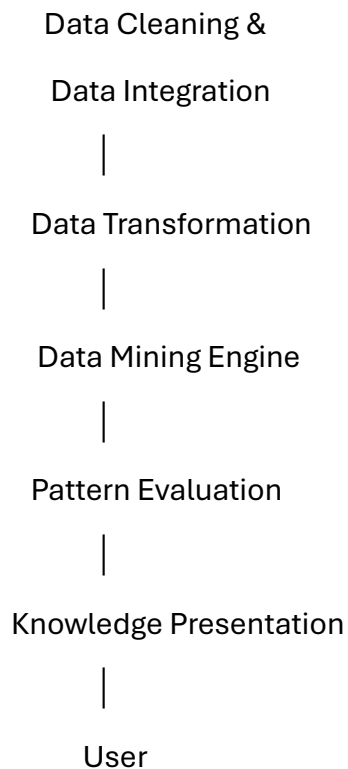
Architecture

Data Sources

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Database/Data Warehouse

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Components

1. Data Sources

- Databases, files, web data.

2. Database/Data Warehouse

- Stores integrated data.

3. Data Cleaning & Integration

- Removes errors and merges multiple sources.

4. Data Transformation

- Normalizes and aggregates data.

5. Data Mining Engine

- Performs classification, clustering, association, etc.

6. Pattern Evaluation

- Selects useful and interesting patterns.

7. Knowledge Presentation

- Displays results using reports, graphs, and visualization.
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Q7. Find Covariance

Given

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 6\}$$

Step 1: Means

$$\bar{X} = 6.2 \quad \bar{Y} = 5.8$$

Step 2

X Y X-6.2 Y-5.8 Product

$$4 \ 8 \ -2.2 \ 2.2 \ -4.84$$

$$5 \ 3 \ -1.2 \ -2.8 \ 3.36$$

$$6 \ 7 \ -0.2 \ 1.2 \ -0.24$$

$$7 \ 5 \ 0.8 \ -0.8 \ -0.64$$

$$9 \ 6 \ 2.8 \ 0.2 \ 0.56$$

$$\text{Sum} = -1.8$$

Sample covariance

$$\text{Cov}(X, Y) = \frac{-1.8}{4} = -0.45$$

Answer: -0.45

The variables have a **weak negative relationship**.

Q8. Describe the major steps in data preprocessing. Propose a preprocessing pipeline.

Major Steps

1. Data Cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction
5. Data Discretization

Systematic Pipeline

Step 1 – Data Cleaning

- Fill missing values.
- Remove duplicates.
- Correct inconsistent formats.

Step 2 – Data Integration

- Merge data from multiple sources.
- Remove redundancy.

Step 3 – Data Transformation

- Normalize data.
- Aggregate attributes.
- Encode categorical values.

Step 4 – Data Reduction

- Remove unnecessary attributes.
- Perform dimensionality reduction.
- Compress large datasets.

Step 5 – Data Discretization

- Convert continuous values into intervals.

Justification:

Cleaning improves quality → Integration combines sources → Transformation prepares data → Reduction speeds mining → Discretization simplifies analysis.

Q9. Equal-Frequency (Equi-Depth) Binning

Given Data (27 values)

13, 15, 16, 16, 19, 20, 20, 21, 22,
22, 25, 25, 25, 25, 30, 33, 33, 35,
35, 35, 35, 36, 40, 45, 46, 52, 70

Divide into **3 equal-frequency bins (9 values each)**.

Bin 1

13,15,16,16,19,20,20,21,22

Mean = **18**

Bin 2

22,25,25,25,25,30,33,33,35

Mean = **28**

Bin 3

35,35,35,36,40,45,46,52,70

Mean = **43.8 ≈ 44**

Smoothed Data

18,18,18,18,18,18,18,18,18,

28,28,28,28,28,28,28,28,28,

44,44,44,44,44,44,44,44,44

This reduces noise while preserving the overall age distribution.

Q10. Explain Data Discretization and compare Equal-Width and Equal-Frequency Binning.

Data Discretization:

It converts continuous numeric values into a limited number of intervals (bins).

Importance

- Reduces noise.
- Simplifies mining algorithms.
- Improves classification.
- Reduces computation time.

Equal-Width Binning

Equal interval size

Easy to implement

Sensitive to outliers

Equal-Frequency Binning

Equal number of records

Better for skewed data

Less affected by outliers

Equal-Width Binning

Equal-Frequency Binning

Uneven number of records Balanced number of records

Example

Data:

10, 20, 30, 40, 50, 60

Equal Width (2 bins):

- Bin 1: 10–35 → 10,20,30
- Bin 2: 36–60 → 40,50,60

Equal Frequency (2 bins):

- Bin 1: 10,20,30
- Bin 2: 40,50,60

Evaluation:

- **Equal-width** is suitable for uniformly distributed data.
- **Equal-frequency** is preferred for skewed datasets because each bin contains the same number of records, leading to more balanced analysis.